

Interview Questions — JavaScript Essentials And Advanced

Total Questions: 52

1. What is JavaScript, and how does it differ when used on the client side versus the server side?

Hint: Think about where the code runs and what it can access.

2. How do you include JavaScript in an HTML page so that it interacts with the web page's content?

Hint: Which HTML tag is used, and where is it usually placed?

3. If you want to quickly test a line of JavaScript code, where can you do this in your browser?

Hint: Most browsers have a built-in tool for developers.

4. Explain the difference between `var`, `let`, and `const` in JavaScript.
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5. What is the difference between primitive and reference data types in JavaScript?

Hint: Think about how values are stored and copied.

6. Give an example of dynamic typing in JavaScript.
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7. What will be printed to the console?

```
let followers = 1500; if (followers > 1000) { console.log('Verified Creator'); } else { console.log('Regular User'); }
```

Hint: Check the value of the variable and the condition.

8. Explain the difference between `==` and `===` in JavaScript.
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9. What is a 'truthy' value in JavaScript? Give one example.

Hint: Not all values need to be true to behave like true in conditions.

10. Write a `for` loop that prints each item in this array to the console:

```
let links = ['Instagram', 'Twitter', 'LinkedIn'];
```
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11. What is the main difference between a `for` loop and a `forEach` loop when working with arrays?
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12. Write a function declaration called `greetUser` that takes a `name` parameter and returns `'Hello, [name]!'`.
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13. What is the difference between a function declaration and a function expression in JavaScript?

Hint: Consider how each can be used before or after its definition.

14. How would you create a button element in JavaScript and set its label and background color using a function?

Hint: Think about using `document.createElement` and setting properties.

15. How do you select an element by its ID in the DOM, and how do you change its text content?

Hint: There are specific DOM methods for selection and updating.

16. Describe how you would add a new link to a list on a web page when a user submits a form.

17. What is the difference between `onclick` and `addEventListener` for handling events in JavaScript?

18. Explain what the event object is and how you might use it in an event handler.

Hint: It provides details about the event that occurred.

19. Write a code snippet that adds a click event listener to a button with the ID `addLinkBtn` and logs 'Button clicked!' when the button is pressed.

20. How would you represent a collection of social links, where each link has a title and a URL, using an array of objects in JavaScript?

Hint: Think about using curly braces inside an array.

21. What is the purpose of `JSON.stringify()` and `JSON.parse()` in JavaScript?

Hint: Consider how data is stored or sent as text.

22. How would you save a user's list of links to `localStorage` and retrieve it later?

Hint: Remember, `localStorage` only stores strings.

23. What are template literals in JavaScript, and how do they differ from traditional string concatenation?

Hint: Look for a special character used to enclose them.

24. Show how you would use destructuring to extract the `title` and `url` from this object: `let link = { title: 'GitHub', url: 'https://github.com' } .`

25. Explain the role of the spread operator (`...`) in copying or merging arrays.

26. What is the difference between blocking and non-blocking code in JavaScript?

Hint: Think about how code execution waits or continues.

27. What is a callback function? Give a simple example using `setTimeout` .

28. Write a Promise that resolves after 2 seconds with the message 'Done!'.

29. How does `fetch()` work in JavaScript to get data from an API? Briefly describe the steps involved.

Hint: Think about the request, response, and parsing process.

30. Show how you would fetch user data from `https://jsonplaceholder.typicode.com/users` and log the result to the console.

31. What is the difference between default exports and named exports in ES6 modules?

Hint: Consider how each is imported.

32. Why is modularizing your JavaScript code (using imports and exports) beneficial for larger applications?

33. How would you move a utility function from one file to another and use it in your main script?

34. How do you declare a class in JavaScript? Write a simple `UserProfile` class with a constructor that takes `name` and `followers` .

Hint: Use the `class` keyword.

35. What is the purpose of `try...catch` in JavaScript, and when would you use it?

36. Write a code snippet that throws a custom error if a provided URL does not start with 'http'.

37. How can AI tools like ChatGPT help you when you encounter an error stack trace in your JavaScript code?

Hint: Think about explanations, suggestions, and verifying fixes.

38. What does the `map()` method do on an array? Give an example.

39. How would you use `filter()` to get only the active links (where `isActive: true`) from an array of link objects?

40. Explain the difference between `find()` and `filter()` when working with arrays.

41. What is event delegation, and why is it useful when creating dynamic lists of elements?

Hint: Think about efficiency and handling events for elements added later.

42. How would you remove a card element from the DOM when its 'X' button is clicked, using event delegation?

43. Name two console methods you would use to debug your JavaScript code and explain what they do.

44. How do you set a breakpoint in the browser's developer tools, and why is this useful for debugging?

45. Describe how you would plan the data flow in a small web app that lets users add, edit, and delete social links.

46. During manual testing, you notice that clicking 'Add' sometimes adds two identical links. What steps would you take to debug and fix this?

Hint: Consider event handlers and form state.

47. How would you validate a URL input in your app and show an error message if it's invalid?

48. In your mini-project, how would you implement a light/dark mode toggle using JavaScript?

49. If you were building your Social Bio Link app, how could you use an AI assistant like ChatGPT or GitHub Copilot to help you write or improve your code? What would you check before using its suggestions?

Hint: Think about code generation, explanations, and reviewing output for accuracy.

50. Integrative: How would you combine ES6 destructuring and advanced array methods to extract all URLs from an array of link objects?

Hint: Consider using `map()` with destructuring in the callback.

51. Integrative: Describe how you would fetch data from an API and then display each item as a new DOM element on the page.

- 52.** Integrative: How would you use classes, event listeners, and localStorage together to allow users to add and persist custom social links in your app?
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